

NETWORK DOCUMENTATION REPORT

Small Network Topology Design, Port Mapping, Service Identification, and Security Assessment.

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Tools Used:

- 1.VMware Workstation
- 2.Windows Server 2022
- 3.Kali Linux
- 4.Ubuntu Linux
- 5.Windows 7
- 6.Command Prompt
- 7.Terminal

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1. Introduction

This report documents a small virtual network environment consisting of four virtual machines: Windows Server 2022, Kali Linux, Ubuntu Linux, and Windows 7. The network topology, device configurations, open ports, services, and potential security weaknesses were analyzed and documented.

2. Objective

The objective of this project is to:

Design a small network topology

Identify network devices

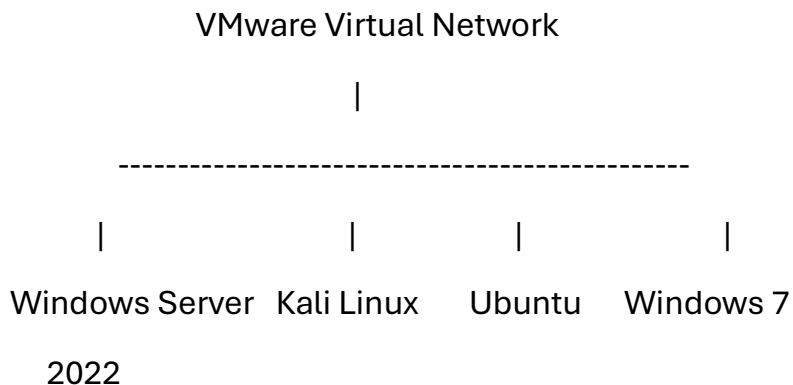
Map open ports and running services

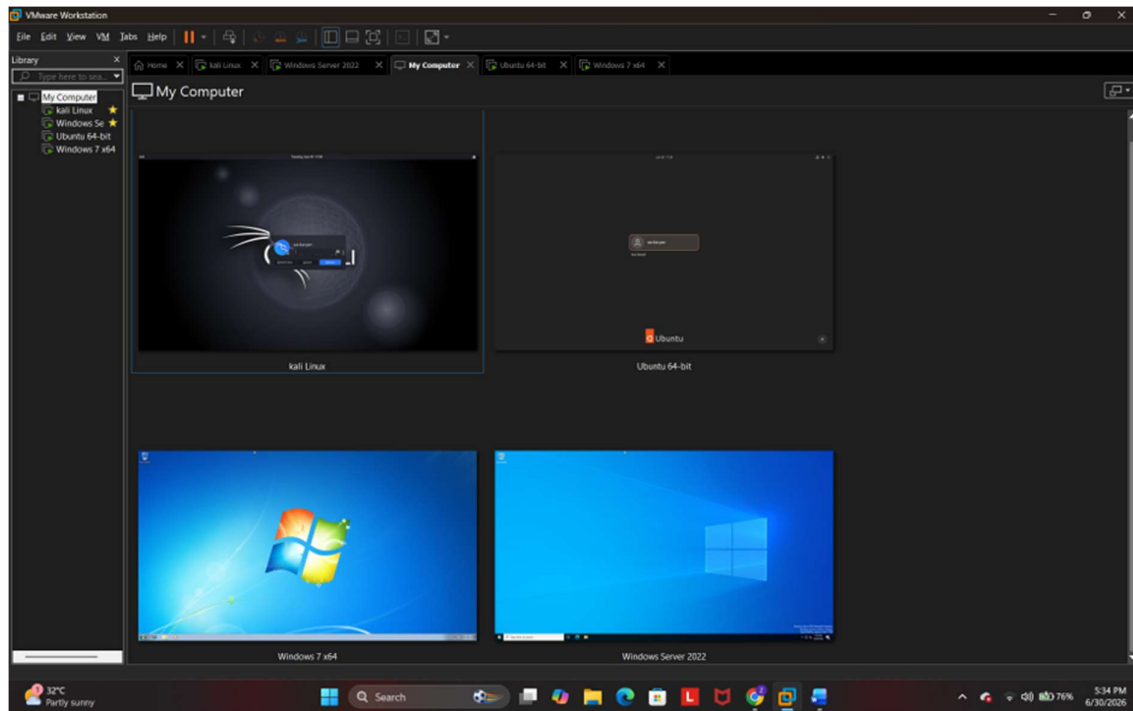
Document potential security weaknesses

Provide security recommendations

3. Network Topology Design

Figure 1: Network Topology Diagram





Description:

Figure 1: VMware Virtual Network Environment

Description:

The VMware Workstation environment contains four virtual machines: Windows Server 2022, Kali Linux, Ubuntu Linux, and Windows 7. These systems are configured within a shared virtual network for communication, testing, and network analysis activities.

4. Device Inventory:

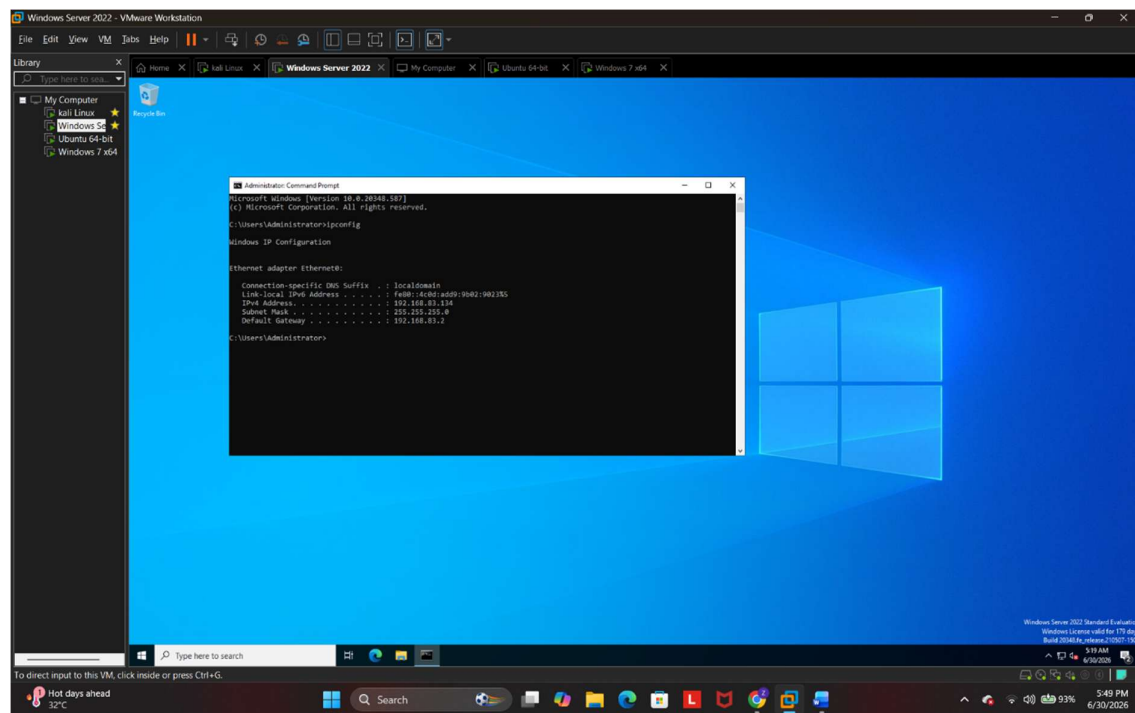
Device Name	Operating System	Purpose
Windows Server	Windows Server 2022	Server
Kali Linux	Kali Linux	Security Testing
Ubuntu	Ubuntu Linux	Linux Workstation
Windows 7	Windows 7	Legacy Client

Description:

The network contains four devices used for administration, testing, and communication purposes.

5. IP Address Configuration

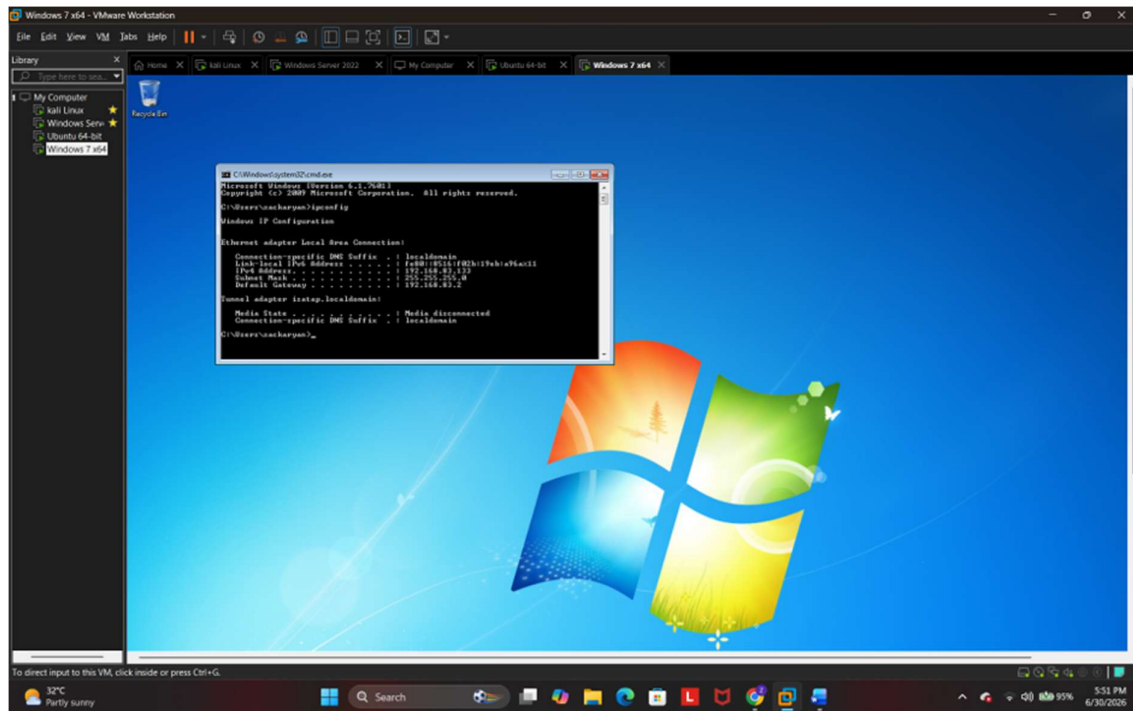
Figure 2: Windows Server 2022 IP Configuration



Description:

IP configuration information obtained using the ipconfig command.

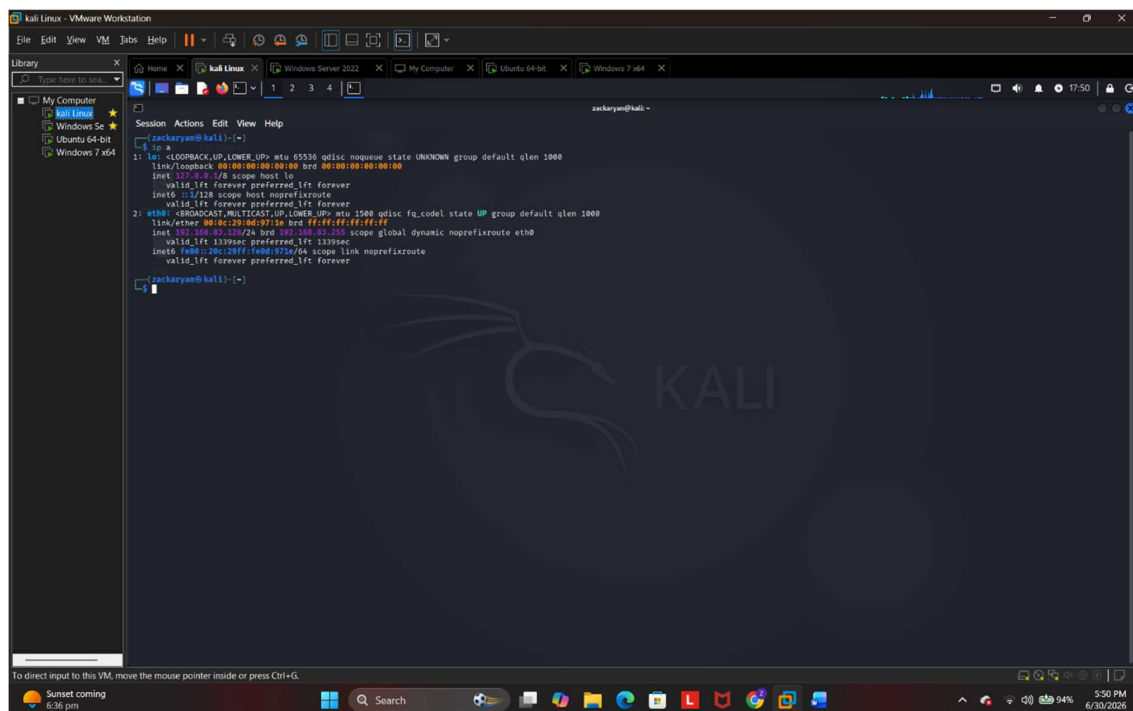
Figure 3: Windows 7 IP Configuration



Description:

IP configuration information obtained using the ipconfig command.

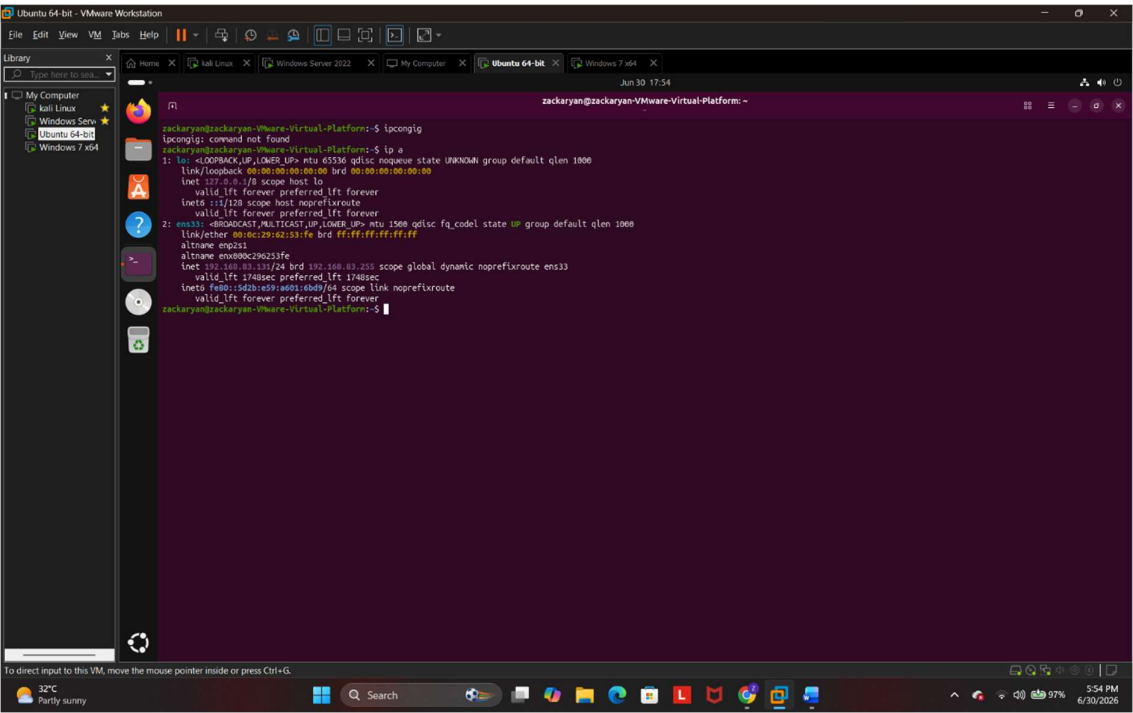
Figure 4: Kali Linux IP Configuration



Description:

IP configuration information obtained using the ip a command.

Figure 5: Ubuntu Linux IP Configuration



Description

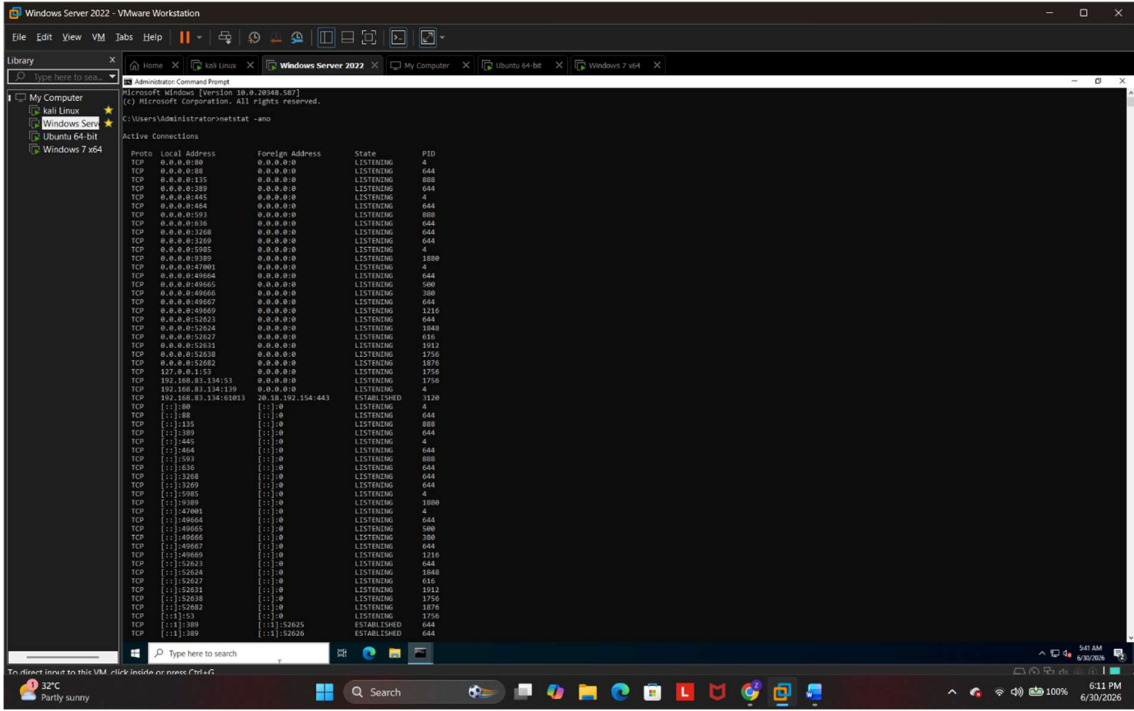
IP configuration information obtained using the ip a command.

DeviceIP	Address
Kali	XXX.XXX.XXX.XXX
Ubuntu	XXX.XXX.XXX.XXX
Windows 7	XXX.XXX.XXX.XXX
Windows Server	XXX.XXX.XXX.XXX

6. Port and Service Mapping:

Figure 6: Windows Server Open Ports

The netstat -ano command was executed on Windows Server 2022 to identify active TCP and UDP connections, listening ports, and associated process IDs. Several network services were observed running on the system.



Command Used

```
netstat -ano
```

Description

The command displays active network connections, listening ports, and associated process identifiers.

Figure 7: Kali Linux Open Ports

Description

The `ss -tln` command was executed on Kali Linux to identify active TCP and UDP listening ports. The output displays network services currently available on the system.



Command Used

`ss -tuln`

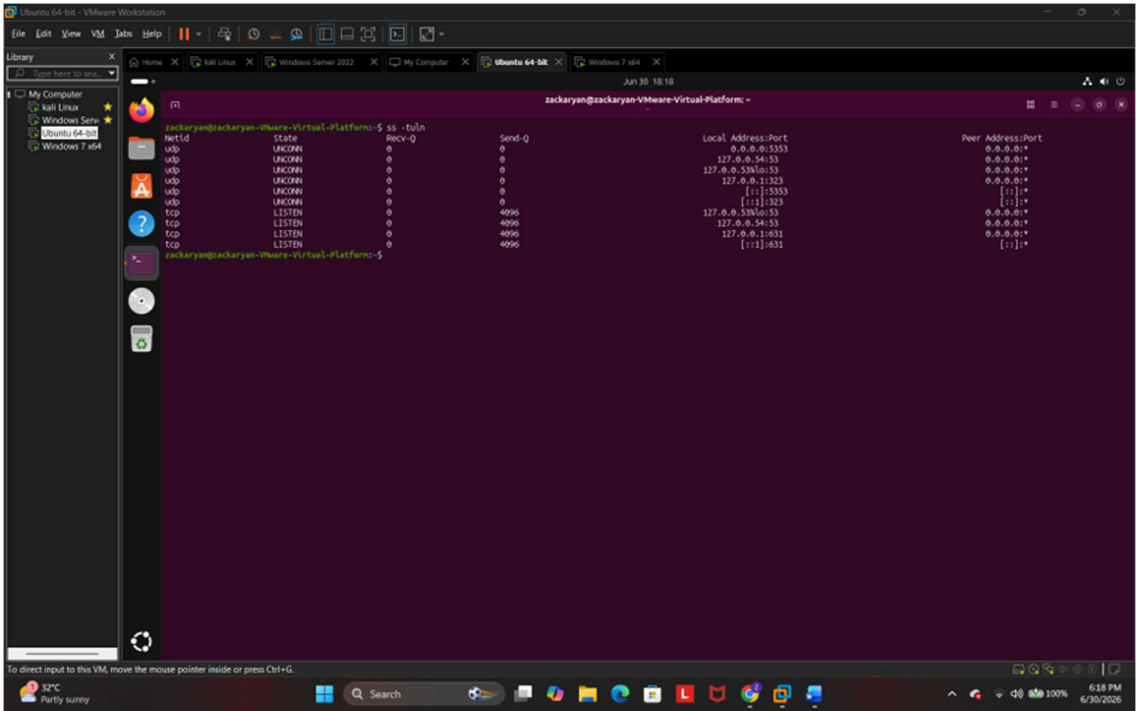
Description

The command displays listening TCP and UDP ports on Kali Linux.

Figure 7: Ubuntu Open Ports

Description

The `ss -tuln` command was executed on Kali Linux to identify active TCP and UDP listening ports. The output displays network services currently available on the system.



Command Used

ss -tuln

Description

The command displays listening TCP and UDP ports on Ubuntu.

Device	Port	Service
Windows Server	3389	RDP
Windows Server	445	SMB
Ubuntu	22	SSH
Windows 7	139	NetBIOS

Port and Service Summary

Port Protocol Service

22	TCP	SSH
53	UDP	DNS
80	TCP	HTTP
443	TCP	HTTPS
3389	TCP	Remote Desktop (RDP)

Description

The identified ports indicate commonly used network services available within the environment.

7. Security Assessment

Potential Weak Point 1 – Legacy Operating System

Windows 7 is no longer supported and does not receive regular security updates. This increases the risk of exploitation through known vulnerabilities.

Potential Weak Point 2 – Open Network Ports

Open ports expose services that may be targeted by attackers if not properly secured.

Potential Weak Point 3 – Weak Authentication

Weak or reused passwords may allow unauthorized access to network resources.

Potential Weak Point 4 – Unencrypted Communication

Services using HTTP instead of HTTPS may expose sensitive information during transmission.

Potential Weak Point 5 – Firewall Misconfiguration

Improper firewall settings can allow unnecessary network access and increase attack surface.

8. Security Recommendations

To improve the security of the network environment:

Keep operating systems updated.

Replace unsupported operating systems such as Windows 7.

Enable host-based firewalls on all systems.

Use strong passwords and account lockout policies.

Disable unnecessary services and ports.

Use encrypted protocols such as HTTPS and SSH.

Regularly monitor network activity and logs.

9. Conclusion

The network documentation project was completed successfully. A virtual network consisting of Windows Server 2022, Kali Linux, Ubuntu Linux, and Windows 7 was documented and analyzed. Device configurations, open ports, and active services were identified. Potential security weaknesses were evaluated, and recommendations were provided to improve the overall security posture of the environment.